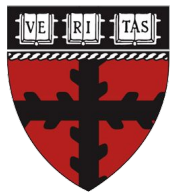
MLOps & LLMOps: Production AI Systems

AC215/E115

Pavlos Protopapas
SEAS/Harvard



Outline

Outline

- About me

Outline

- About me
- Why take this course in the era of generative AI?

What is in this video

- About me
- Why take this course in the era of generative AI?
- What the course is about, and what you need coming in

About me

Pavlos Protopapas

- Scientific Program Director, Harvard SEAS
- PhD in physics, University of Pennsylvania



About me

Pavlos Protopapas

- Scientific Program Director, Harvard SEAS
- PhD in physics, University of Pennsylvania

Research: machine learning + the physical sciences

- PINNs: solving PDEs with neural networks
- Language models
- Cybersecurity for edge devices
- Foundation models in astronomy



About me

Teaching

CS1090A/B
Introduction to Data Science

AC215
This Course

ACTIVE

HarvardX
Online Courses



About me

Hobbies

- Cheese — eating it, and making it
- Classical music — invite me to your performances
- Sports — football (soccer) and cycling



All staff:
https://harvard-iacs.github.io/2026-AC215/staff/

In the era of generative AI,

if agents can build all of this…

why take this course?

What we expect from you

You can build the entire system without generative AI

- Slower? Yes. Documentation and Stack Overflow? Yes.
- But you know the steps, what has to be done, and how to stitch it together yourself

AI can help you build faster. This course makes sure you understand what you are building.

What we assess

- **Understanding** — quizzes test the core concepts and technical decisions
- **Engineering** — well-structured code, sound engineering practices
- **Systems** — coherent, working pipelines, not disconnected pieces
- **Communication** — explain, demonstrate, and present every part of your system

Why Learn the Steps?

Autonomous Fails

Fully autonomous agents are not ready for prime time.

Today's autonomous systems deliver unpredictable results and are incredibly expensive to fix when they wander off track.

Directed Execution

Human guidance is the secret to high performance.

Collaborative AI works. Guiding agents step-by-step unlocks ~90% efficiency gains.

e.g., Outline → Slides → Script

The Expertise Gap

AI is a multiplier, not a baseline creator.

AI agents amplify potential, but only if you already deeply understand the underlying steps and workflows yourself.

This course teaches you those steps.

The cycling coach

- Sooner or later, you will be **guiding agents** — or **guiding people**
- Imagine a **cycling coach** who has **never ridden a bike**
- **Knows FTP, power output**, oxygen levels, glucose levels…but has **never felt a long climb**
- **You cannot coach a sport you have never played**



In this course, you play the sport

What this course is about

Building a model is only half the journey

- The other half: a system that is reliable, scalable, and usable
- Full lifecycle of ML systems — with a strong emphasis on LLMs

The goal: design, build, and deploy a complete AI system, end to end

Course Logistics

Logistics Video

Comprehensive pre-recorded walkthrough.

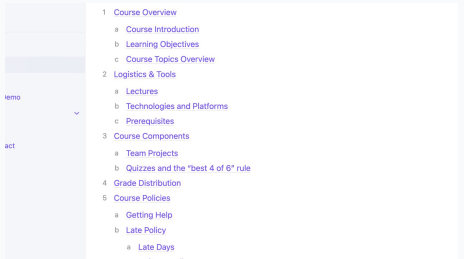
Watch our video to understand how the course runs day-to-day.



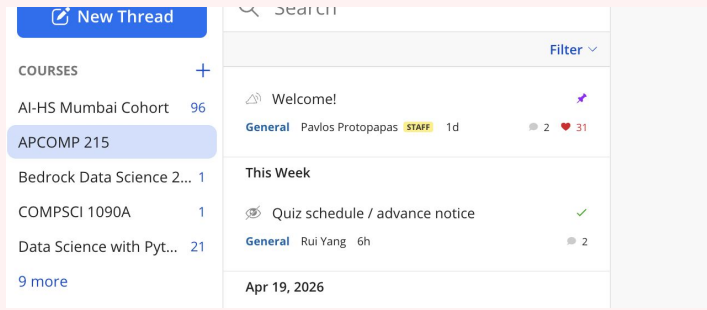
Course Syllabus

Policies, grading, & timelines.

The ultimate source of truth for assignments and expectations.



Please feel free to ask any questions in the community forum or during office hours!



Course Assessment

An important component you need to know is the assessment.
These two parts are critical to your success!

Quizzes

Conceptual & Practical Coding

- **Format:** Short (20–30 mins) during class time
- **Content:** Mix of theory and short coding tasks
- **Scope:** Recent material; tests practical understanding
- **On-campus:** In person only (no online, no make-ups)
- **DCE students:** Conducted online
- **Test Quiz:** Prepared to get a flavor on Edstem (https://edstem.org/)

Milestone Evaluations

Hands-on System Performance

- **Focus:** End-to-end ML pipelines evaluating quality, scalability, and reliability
- **MS2–MS4:** 15-minute team presentation in weekly TF meetings
- **Guidelines:** Detailed criteria are posted on Edstem (https://edstem.org/us/courses/97373/lessons/180730/slides/1062993)

Is This Course For You?

Now that you know the **rigor** of our quizzes, the **depth** of our milestones, and the **full-stack execution** required...

You have a critical decision to make.

No pressure, but your keyboard is waiting. 😉

Decision time: Are you ready to play the sport, or stay on the sidelines?